

Keisuke Kwong Fall 2026 First Report

Keywords

racquetball launcher (formerly skeeball launcher), waterproofing, motor enclosures, SolidWorks

Abstract

This report summarizes my work on the racquetball launcher for the first half of the fall 2026 semester (up to September 27th). In particular, this report focuses on waterproofing the racquetball launcher system, particularly the motor enclosure. Waterproofing is vital for this system because it sits on the deck of the boat, so it is prone to splashes from waves and impact from rain. The goal is for the enclosures to be able to withstand soaking in water for up to 12 hours without water entering the enclosure. I have created several iterations of enclosure designs for the various points of entry that require sealing, which include rotary O-rings for shafts, face O-ring seals, and wire glands. Currently, I have been able to test each type of waterproofing, so I am ready to test the entire assembled enclosure for waterproofing. After this, I can complete the racquetball launcher assembly and begin range testing before the final design review.

Technical Writeup

Current Solution

The racquetball launcher is one of the main robotic components for our autonomous boat. It is used to deliver racquetballs to nearby vessels and consists of two subsystems:

1. Feeder Mechanism

- The circular base holds the three racquetballs.
- The rotating plate dispenses the balls, one at a time, into the launching mechanism.
- A servo with a 180° range of motion controls the rotation and release of balls.

2. Launching Mechanism

- A rubber flywheel accelerates the ball through a ramp.
- A 12V DC motor spins the wheel.

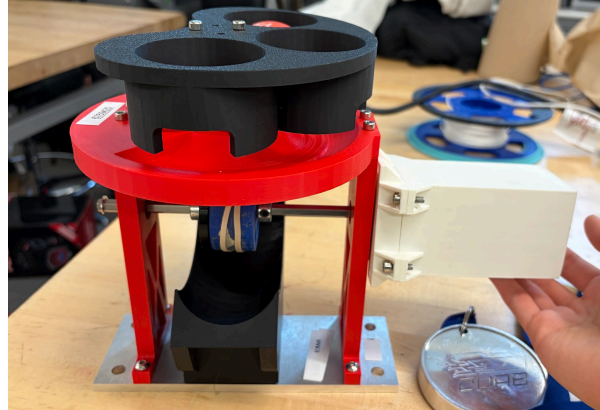


Figure 1. Racquetball launcher

More specifically, the main components of the system are as follows:

- Aluminum **base plate**
- 3D-printed PLA **ramp** with acrylic **side plates**
- **Rubber flywheels** with 3D-printed TPU **wheel attachment**
- 3D-printed PLA **motor enclosure** around the **12V Brushed DC Motor**, which is connected to the **hex shaft**
- **HS-311 servo** contained within the **servo enclosure** attached to the **feeder base** and **spinning top** (note: servo enclosure, feeder base, and spinning top are 3D-printed PLA)
- 3D-printed PLA **vertical supports** that support the feeder and fasten to the base plate

Overview

The racquetball launcher assembly was mostly designed and assembled by the end of spring 2026. However, as of the start of fall 2026, the servo enclosure had waterproofing issues with the face seal gland, and the motor enclosure had major design flaws regarding assembly and had not yet been tested for waterproofing. My main focus for the start of this semester was to completely revise the motor enclosure, fabricate a working prototype, and begin its waterproofing validation testing.

Table 1 summarizes the key design requirements for the servo and motor enclosures:

Table 1. Key Design Requirements

Waterproof	Enclosures must protect the servo and motor from water damage (e.g., waves, splashes) during operation, and metal parts should be corrosion-resistant.
Assembly	Enclosures must secure the servo and motor in place, be able to be assembled reasonably quickly, and must not interfere with the larger assembly.
Size	Enclosures must be large enough to house the servo and motor, but their footprint should be minimized to fit in the limited space on the boat's deck.

Additionally, the robotics subteam decided to make a CAD software switch to SolidWorks starting this semester, so I rebuilt the entire racquetball launcher assembly in SolidWorks while simultaneously learning the software and troubleshooting 3DEXPERIENCE.

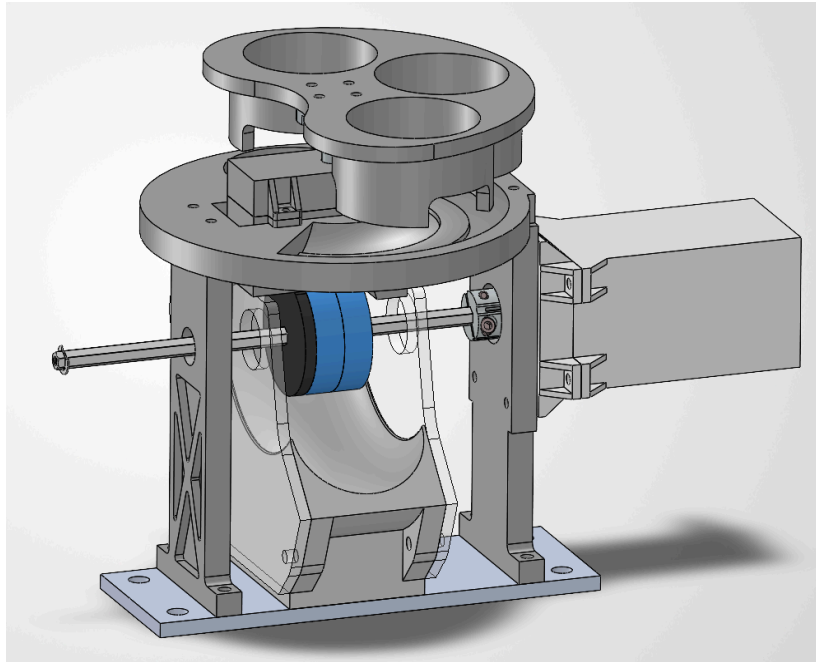


Figure 2. Racquetball launcher in SolidWorks

Currently, the final designs for the servo and motor enclosures are ready to be fabricated and tested. Since they have already been tested separately for waterproofing (detailed in the next section), I expect this testing to be completed in 1-2 weeks, which will allow us to assemble the full system and prepare for range testing in the next month.

Design Details

As detailed earlier, both the servo and motor enclosures are made out of PLA plastic, as they are 3D-printed to be custom-sized to fit our needs and the servo and motors we use.

The first major change I made was shifting from the angled motor enclosure design from spring 2026 back to a straight motor design. There were two reasons for this switch. First, I wanted to guarantee a working prototype by the final design review (FDR), and adding the extra complexity of 90-degree miter gears seemed likely to prolong this process. Second, and more importantly, there were major manufacturing and assembly flaws with this design. The angled design involved placing heat-set inserts for screws in locations difficult to access, and the enclosure also completely blocked the screw holes for fastening the entire racquetball launcher assembly to the boat. In other words, the motor would have to be attached *after* first fastening the assembly to the boat, which was highly undesirable.

With this new design, three main points of the motor enclosure require waterproofing:

1. Entrance for motor wires to provide electrical power to the motor
2. Exit for the hex shaft to transfer mechanical power out of the motor
3. Gap that results when assembling the two halves of the motor enclosure

I will detail my work on each of these points below.

1. Wire gland for motor wires

There are six wires that must exit the motor enclosure: one 12V, one GND, and four encoder wires. The details of the electrical components can be found on the [“For Electrical and Software” Confluence page for the racquetball launcher](#). The 12V and GND wires are of a relatively low gauge, creating gaps between the wires through which water can seep. Accordingly, simply squeezing the wires together with an off-the-shelf wire gland was insufficient.

To solve this problem, I lined the gaps between the wires with Oatey plumber’s putty and then used heat shrink around the bundle of wires. Then, squeezing this bundle tightly with an M16 wire gland was sufficient to stop water from flowing through. See Figure 3 in the appendix for a close-up of the wire gland.

2. Hex shaft rotary O-ring seal

The major difficulty of waterproofing the hex shaft is the fact that the shaft is hex, while rotary O-rings only work on round objects. So, I designed an adapter that attaches around the hex shaft and has grooves for round O-rings around the outside. See Figure 4 in the appendix for a CAD image of the hex adapter.

The part is 3D-printed out of TPU to take advantage of the flexible nature of TPU to reduce the gaps between it and the hex shaft. However, this alone is not enough to prevent water from leaking inside, so we once again used Oatey plumber’s putty between the hex shaft and the hex adapter. Note that this works because the hex adapter is fixed to the hex shaft.

For the rotary O-rings on the outside of the adapter, we used two O-rings because using only one allowed the hex adapter to wiggle inside the motor enclosure. Two O-rings made the adapter more secure to the enclosure. To determine the dimensions of the gap, I tested several designs of the adapter until I found one that fit in the motor enclosure hole and matched the size of the 012 O-ring. See Figure 5 in the appendix to see the printed hex adapter (green) in the motor enclosure.

3. Face seal O-ring

The face seal took the longest to waterproof. The reason for this was that it took many iterations to determine the correct dimensions of the groove to house the 1/16" nominal diameter O-ring. We fastened the two halves of the enclosure with M4 screws on tabs that protruded from each part, and strengthened the tabs with ribs, similar to those detailed in [my report from spring 2026](#).

In the end, we used a design that incorporates a female groove on the face of the left half of the enclosure and a male groove on the face of the right half of the enclosure to apply greater pressure onto the O-ring. See Figures 6-7 for the CAD model of both halves of the enclosure, Figure 8 for an example test part for this face O-ring, and Figure 9 for the final dimensions of the groove.

See Figure 10 for the completed motor enclosure CAD and Figure 11 for a preliminary CAD (this prototype is mostly identical to the CAD from Figure 10, except it does not include the male part for the face seal).

Although I was able to verify waterproofing abilities for parts 1-3 independently, I have not yet fabricated and tested the fully updated motor enclosure. Also, it should be noted that my results from part 3 will translate directly to the servo enclosure, as the same O-ring is used in the face seal there. The remainder of the servo enclosure waterproofing has been completed in spring 2026, so the servo enclosure is at the same position as the motor enclosure—ready for final waterproofing testing.

I plan to conduct the waterproofing testing by October 4, which will allow us to collect updated measurements for the range of the fully assembled launcher before Final Design Reviews on October 18. I hope to have the completed assembly ready for pond/pool testing by November.

Design Archive

See Table 2 for the bill of materials for the racquetball launcher system.

Table 2. Bill of Materials

Subassembly	Item	Quantity
Main assembly	Aluminum 6061 Base Plate, 1/4" thickness stock	1
	Ramp, PLA	1
	Acrylic Sheets 12" x 12" x 3/16" (link)	2
	8mm Bore Face Thru-Hole Pillow Block (32mm x 32mm Mounting Pattern) (link)	2
	Vertical supports, PLA	2
	18-8 Stainless Steel Tapered Heat-Set Insert for Plastic, M4 x 0.7 mm Thread Size, 7.92 mm Installed Length (link)	4
	18-8 Stainless Steel Tapered Heat-Set Insert for Plastic, M5 x 0.8 mm Thread Size, 11.1 mm Installed Length (link)	4
	3 in. Compliant Wheel, 3/8 in. Hex Bore, 50A Durometer (link)	2
	Wheel attachment, TPU	1
	uxcell SF688ZZ Flanged Ball Bearing 8x16x5mm Double Shielded Stainless Steel Bearings 4pcs (link)	1
	4007 Series Hyper Coupler (8mm REX™ Bore to 8mm REX™ Bore) (link)	1
	2920 Series Steel Set-Screw Collar (8mm REX™ Bore) (link)	3
	8mm REX® Shaft with E-Clip (Stainless Steel, 192mm Length) (link)	1
	Feeder subassembly	1
Feeder subassembly	Feeder base, PLA	1
	Spinning top, PLA	1
	HS-311 Servo (link)	1
	ServoCity 1910 Series Servo Hub-Shaft (24 Tooth Spline, 10mm Shaft Diameter, 33mm Length) (link)	1
	Servo enclosure top, PLA	1
	Servo enclosure bottom, PLA	1
	X-Profile Oil-Resistant Buna-N O-Ring, 1/16 Fractional Width, Dash Number 012 (link)	1
	.070" (1/16" Nominal) Buna-N O-Ring Cord Stock, 70A Durometer, 0.070" Thickness, 10' Piece, Black (link)	1
	LOKMAN Cable Gland Plastic Waterproof Adjustable 12 mm Cable Connectors with Gaskets (link)	1

	18-8 Stainless Steel Tapered Heat-Set Insert for Plastic, M3 x 0.5 mm Thread Size, 5.56 mm Installed Length (link)	4
Motor subassembly	goBILDA 5000 Series 12VDC Motor (link)	1
	Encoder Breakout Cable Female (4-Pos JST XH [MH-FC] to 4 x 1-Pos TJC8 [MH-FC], 300mm Length) (link)	1
	Motor enclosure left half, PLA	1
	Motor enclosure right half, PLA	1
	Hex adapter, TPU	1
	X-Profile Oil-Resistant Buna-N O-Ring, 1/16 Fractional Width, Dash Number 012 (link)	2
	.070" (1/16" Nominal) Buna-N O-Ring Cord Stock, 70A Durometer, 0.070" Thickness, 10' Piece, Black (link)	1
	LOKMAN Cable Gland Plastic Waterproof Adjustable 16 mm Cable Connectors with Gaskets (link)	1
	Heat shrink tubing	1
	Oatey® Stain-Free Plumber's Putty (link)	1
	18-8 Stainless Steel Tapered Heat-Set Insert for Plastic, M4 x 0.7 mm Thread Size, 7.92 mm Installed Length (link)	4

Reflection

One of the major difficulties of the racquetball launcher waterproofing was complications created by the hex shaft of the motor, which led to the current solution using the hex adapter. However, I believe that I would have been able to solve this issue much earlier, perhaps last spring, if I had decided to go with the simpler straight motor design from the start, rather than pursuing the angled enclosure route, trying to slightly lower the center of gravity without realizing the design's major flaw for assembly. In the future, I should spend more time analyzing new ideas and designs before jumping right into them, as this ultimately would save a lot of time.

On the topic of time, the only time constraint I faced was trying to develop a working prototype in order to collect testing data to present at our final design review (FDR) on October 18th. At my current stage, I have already been able to conduct several iterations of waterproofing testing, and am in a good position to begin range testing before FDR.

Appendix

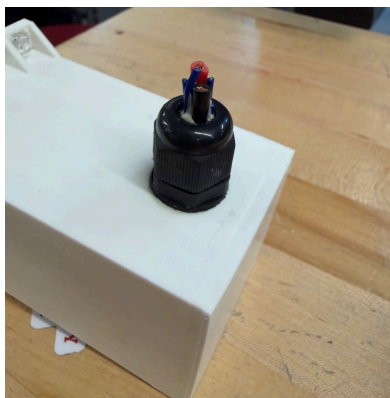


Figure 3. Wire gland seal

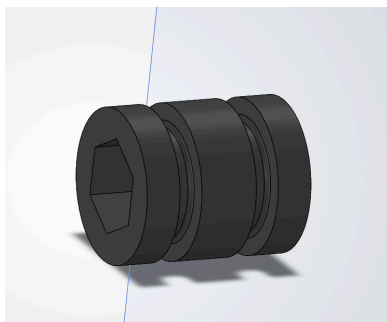


Figure 4. Hex adapter

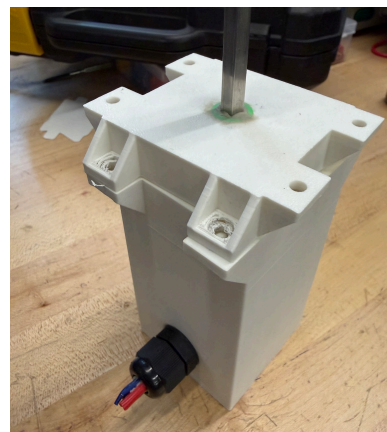


Figure 5. Motor enclosure

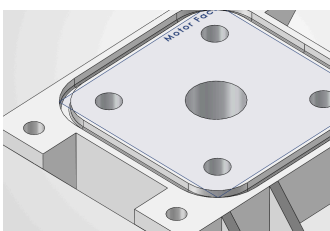


Figure 6. Left half of enclosure CAD

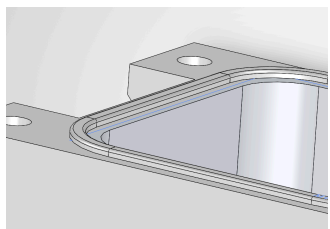


Figure 7. Right half of enclosure CAD



Figure 8. Final test part with male/female groove

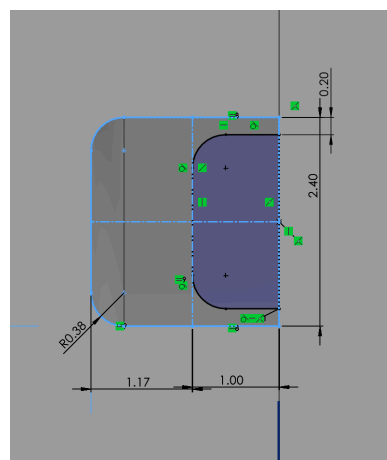


Figure 9. Dimensions of face seal O-ring groove (male and female parts) (mm)

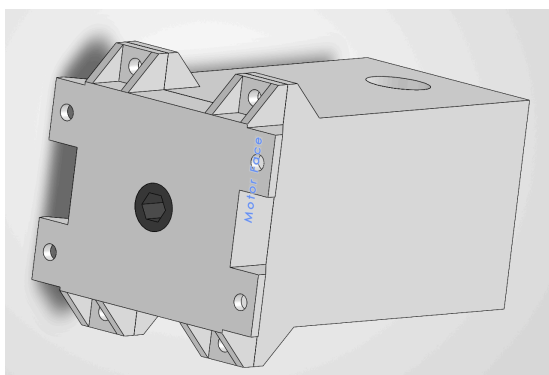


Figure 10. Full motor enclosure CAD

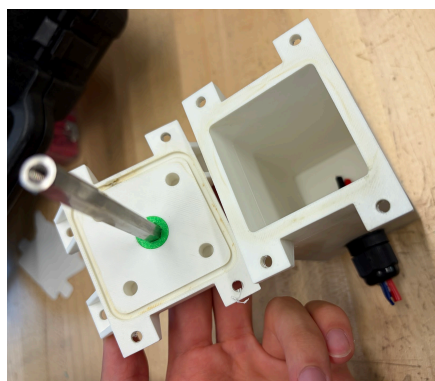


Figure 11. Preliminary motor enclosure